ETHER

Build Instructions

V1.1

Building the Ether

Thank you for purchasing the Ether DIY Kit.

This kit is designed for through-hole assembly: all SMD components are already pre-soldered. The project is well within reach for anyone with basic soldering experience, take your time and follow the instructions step by step.

Warranty

BLACK NOISE warrants the contents of this kit to be free of defects in materials or workmanship and to conform to specifications at the time of shipment for a period of two years from the date of purchase.

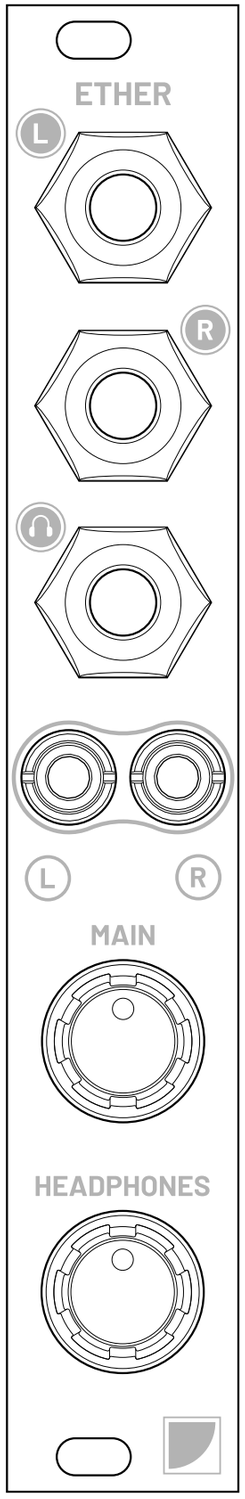
BLACK NOISE cannot be held responsible for damage resulting from incorrect assembly, soldering errors, or faulty handling during the build process.

Due to the difficulty of troubleshooting or debugging a module remotely, we do not provide assistance.

If your module is not functional due to an error during assembly, you can use our repair service: Magic Smoke Club.

For more information, please visit

www.blacknoisemodular.com



Check the components list

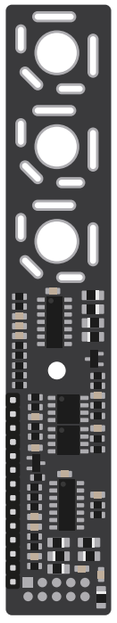
Before starting the build, make sure you have all the components listed in the kit.

Keep each bag's components separated, as they will be used in sequence during the build.

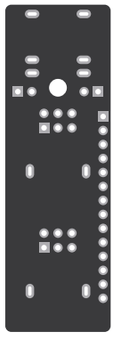
01 Faceplate



02 PCB Control Board



03 PCB Main Board



M3 Hex Screw



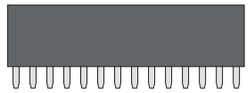
A1 **x2**

M3 Spacer



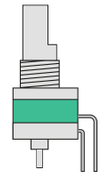
A2 **x1**

Female PinHeader 14x1



A3 * **x1**

Pot. Stereo B100K-D



A4 **x2**

LED THT



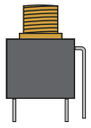
A5 **x2**

LED Spacer



A6 **x2**

Mono Jack Conn.



A7 **x2**

Jack Nut



A8 **x2**

Pot Nut Black



A9 **x2**

Power PinHeader 5x2



A10 * **x1**

TRS 6.5 Jack Conn.



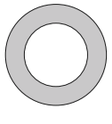
A11 **x3**

6.5 Jack Nut Black



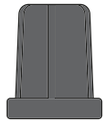
A12 **x3**

6.5 Jack Washer



A13 **x3**

Knob



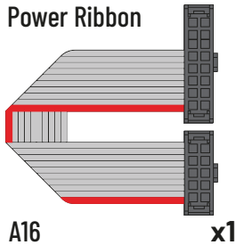
A14 **x2**

Knob Cap



A15 **x2**

Power Ribbon



A16 **x1**

*** First batch white PCB kit : already pre-soldered**

Install spacer on the back of the small board

Take the small PCB.

Insert the 3 mm screw from the front side.

On the back side, fix the metal spacer onto the screw.

4

M3 Hex Screw

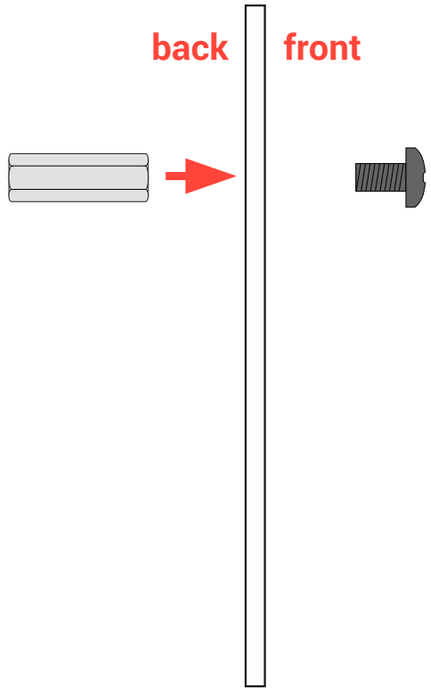


A1 **x1**

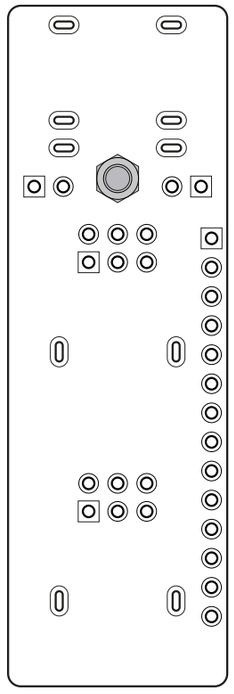
M3 Spacer



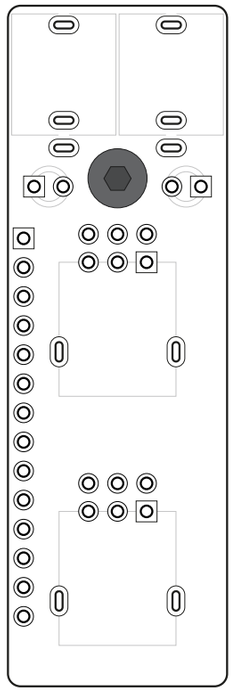
A2 **x1**



Left Side



Back Side



Front Side

Solder the connector

First batch white PCB kit :

already pre-soldered, skip this step.

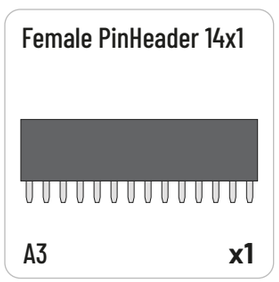
Take the female 14-pin header.

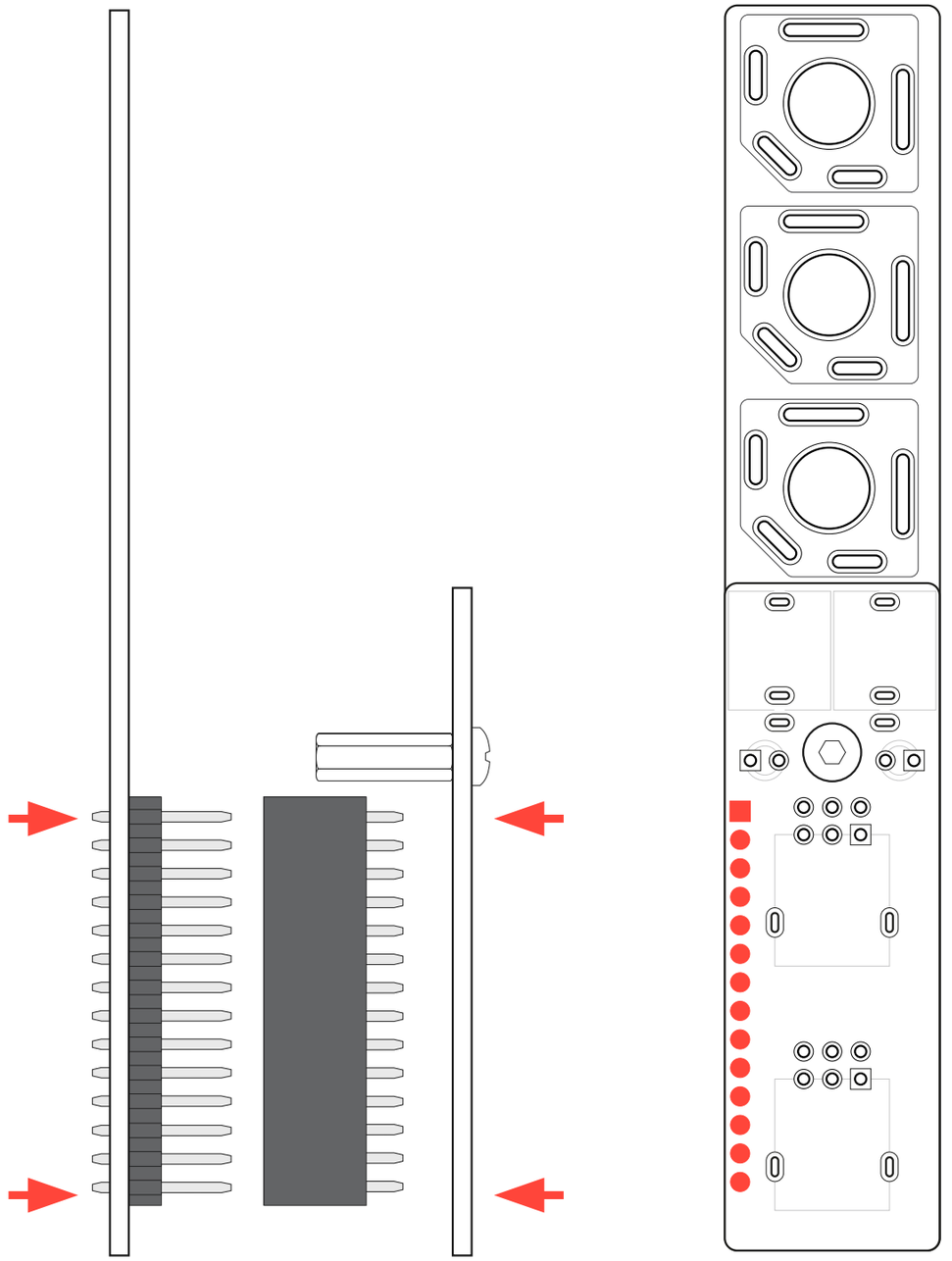
Insert it onto main PCB’s pre-soldered male 14-pin header.

Place the small PCB on top, so the female header’s pins pass through its holes. The spacer now rests on the work surface, acting as a support leg.

Solder the female header’s pins from the front side of the small PCB, where they come through.

Check that all pins are soldered and that there are no solder bridges. Then clean the flux residue if needed.





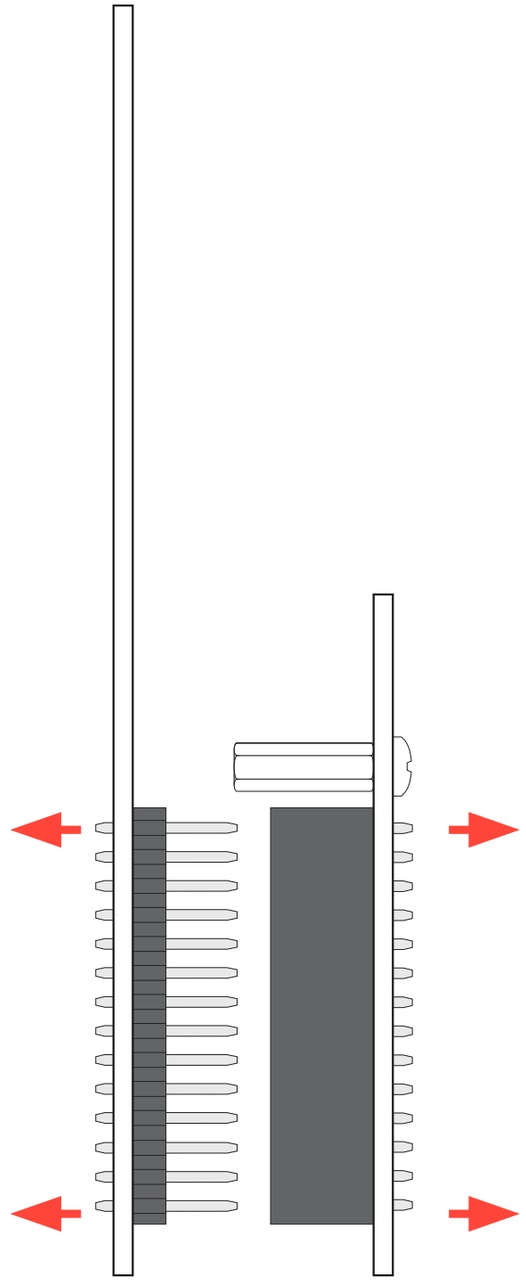
Left Side *Front Side*

Separate the boards

First batch white PCB kit :

already pre-soldered, skip this step.

Separate the two boards so the female header comes away with the small PCB.



Left Side

Insert the 2 potentiometers

Do not solder yet!

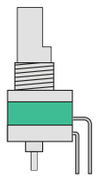
Take the small PCB

Lightly bend the side legs inward so they slide in more easily.

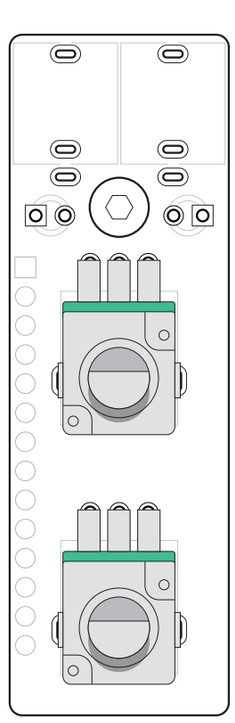
Insert the 6 main pins first.

Insert the side legs one after the other. Take your time and be patient.

Pot. Stereo B100K-D



A4 **x2**



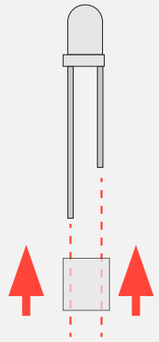
Front Side

Insert the LEDs and LED spacers

Do not solder yet!

Insert the two LEDs into the provided plastic LED spacers.

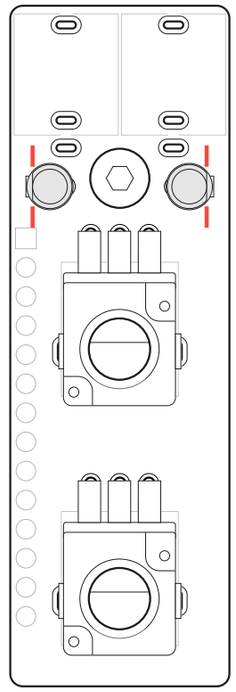
1. Make sure the LEDs sit flush in the spacers.



2. Observe the polarity: the long leg (anode) goes in the round hole, the short leg (cathode, flat side) goes in the square hole. Make sure of the correct orientation by looking at the PCB from the back side and confirming that the short leg comes out of the square pad.



LED THT	LED Spacer
	
A5 **x2**	A6 **x2**



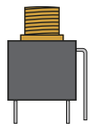
Front Side

Insert the 2 minijack sockets

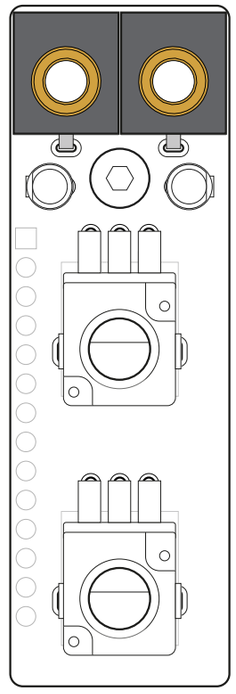
Do not solder yet!

Place the two 3.5 mm jack sockets.

Mono Jack Conn.



A7 **x2**

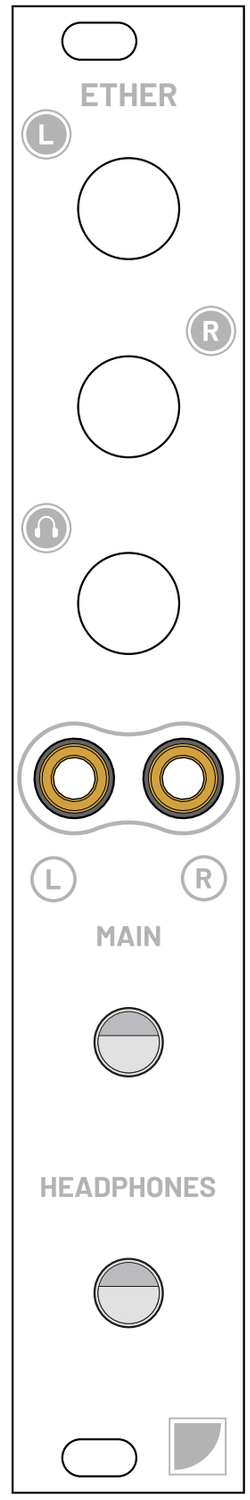


Front Side

Place the faceplate

Carefully slide the faceplate over all the components.

Ensure it doesn’t get stuck on any potentiometer shafts, or jacks.

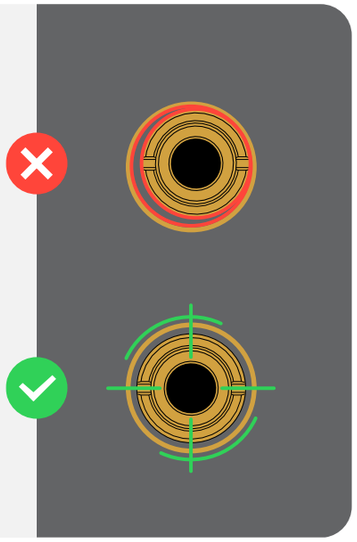


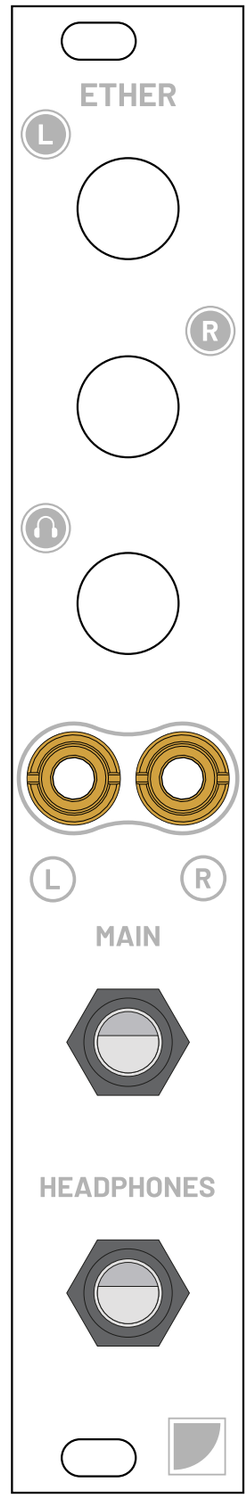
Front Side

Secure the faceplate

Place the potentiometer and jack nuts.

When a circle is printed around a jack nut on the faceplate, make sure the nut is centered within that circle before tightening it.





Front Side

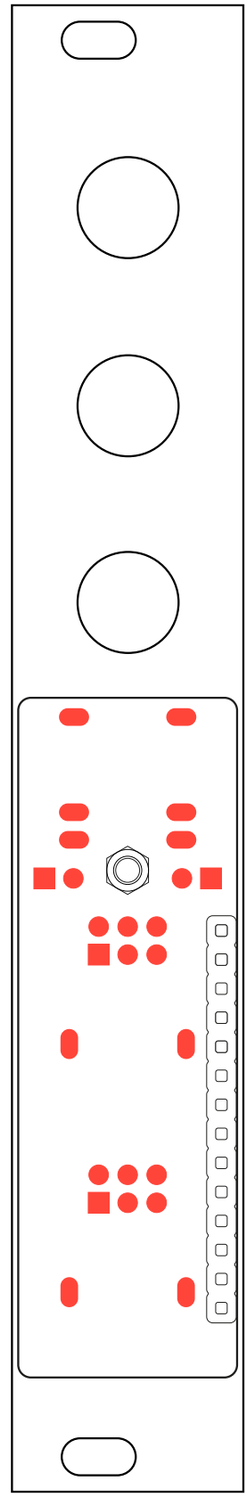
Jack Nut	Pot Nut Black
A8 **x2**	A9 **x2**

Solder the components

Potentiometers, LEDs, and jacks can now be soldered.

Check that all pins are soldered and that there are no solder bridges. Then clean the flux residue if needed.

After soldering, cut the excess legs of the LEDs flush with the PCB.



Back Side

Solder the power connector

First batch white PCB kit :

already pre-soldered, skip this step.

Take the main board.

Place the power connector on the back of the PCB, with its long legs facing away from the pre-soldered 14-pin connector.

Solder one leg first to check the connector is properly positioned and aligned, then solder the remaining legs.

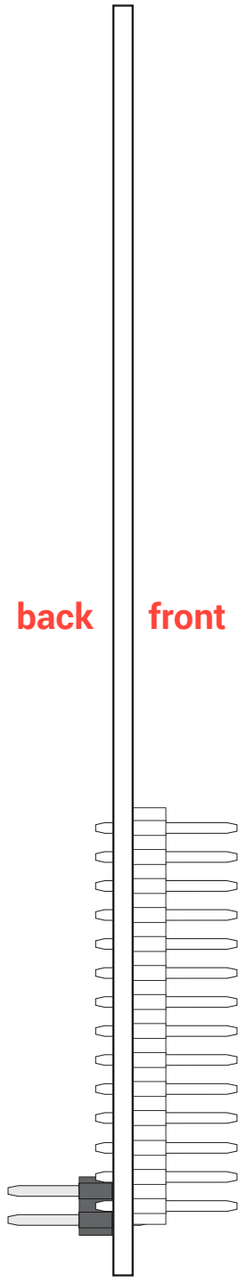
Check that all pins are soldered and that there are no solder bridges,

Clean the flux residue if needed.

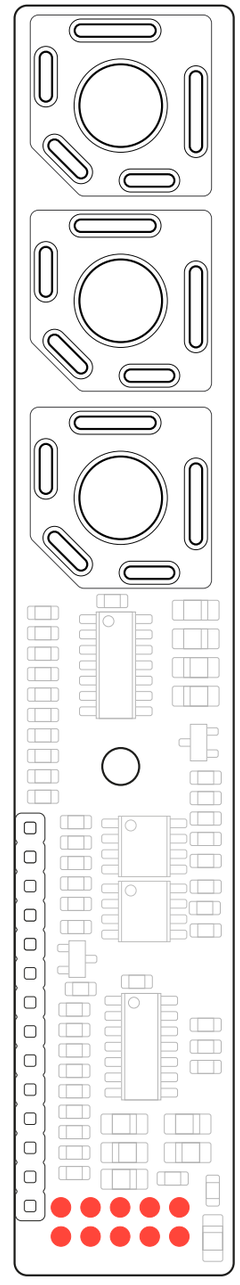
Power PinHeader 5x2



A10 **x1**



Left Side

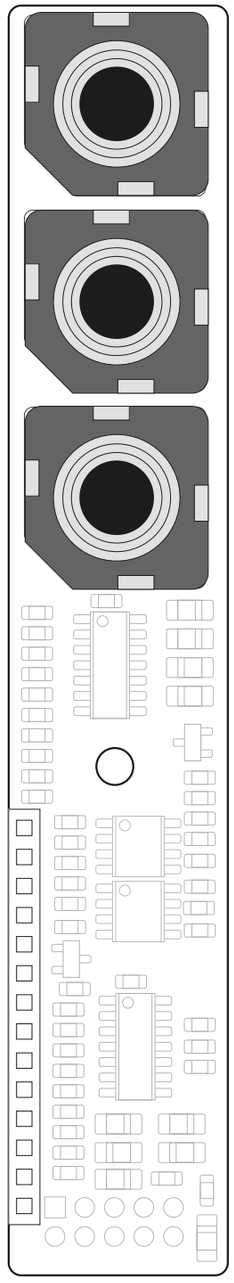


Front Side

Insert the TRS jack sockets

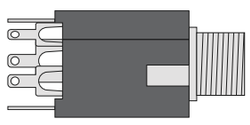
Do not solder yet!

Insert the three 6.35 mm TRS jack sockets.



Front Side

TRS 6.5 Jack Conn.



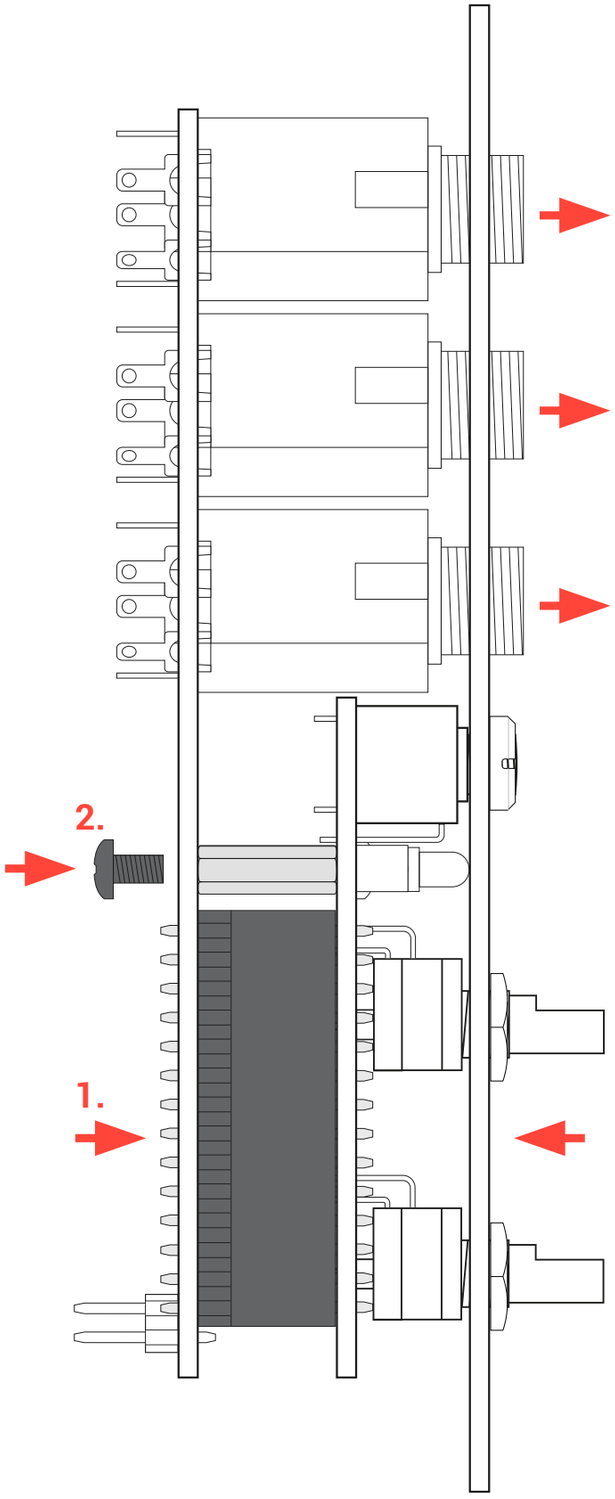
A11 **x3**

Connect the main board to the control board

Insert the 14-pin male header into the 14-pin female header to connect the two boards together (**1**).

Press gently until both boards sit flush.

On the back side, insert the second screw into the spacer to tighten the build (**2**).



Left Side

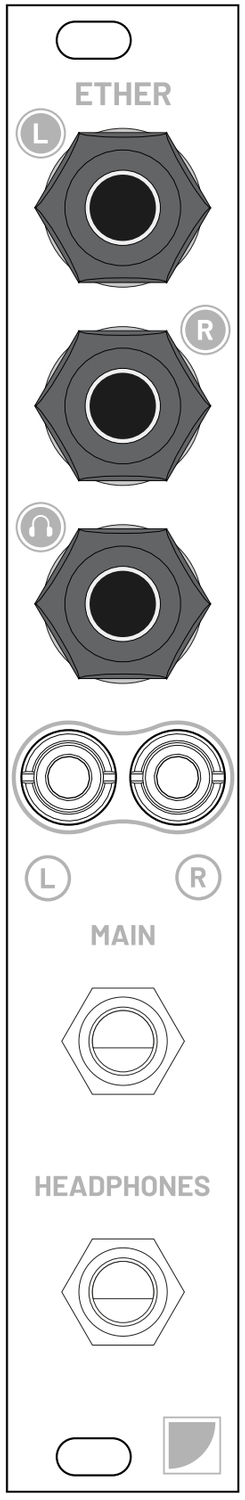
M3 Hex Screw



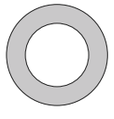
A1 **x1**

Install the TRS jack nuts

Place the jack washers and nuts on the 6.5 Jack sockets.



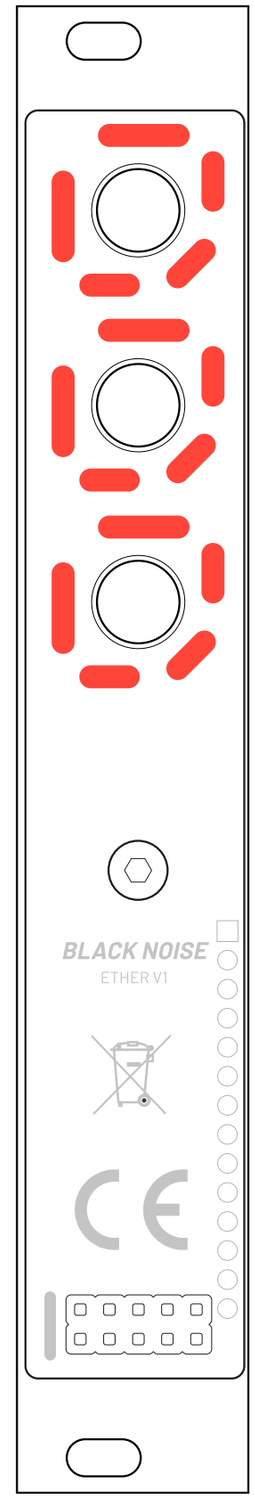
Front Side

6.5 Jack Nut Black		6.5 Jack Washer	
			
A12	**x3**	A13	**x3**

Solder the TRS jack sockets

The TRS jacks can now be soldered.

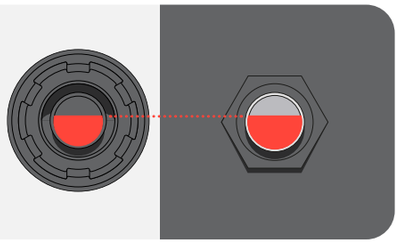
Clean the flux residue if needed.



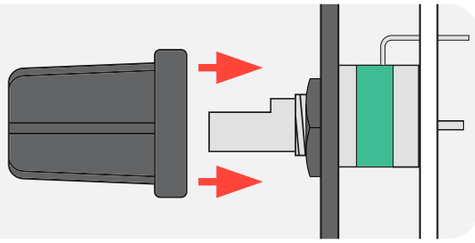
Back Side

Install the knobs

1. Align the D-shape inside the knob with the D-shape of the potentiometer shaft.



2. Place the knob on it potentiometer.



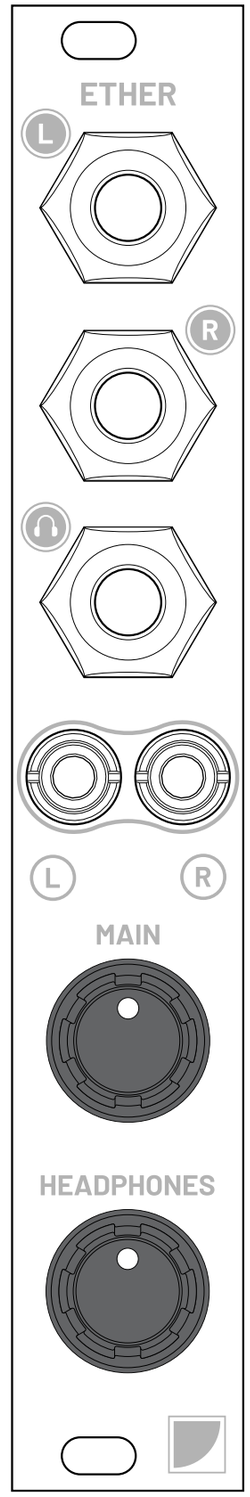
3. Turn the knob fully counter-clockwise (minimum position) so it stays stable.



4. Place the cap so the white dot marks the starting position. Align the dot with the knob's slit. take your time and be gentle.



Knob	Knob Cap
	
A14 **x2**	A15 **x2**



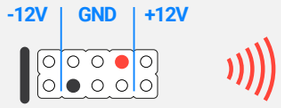
Front Side

Test for short circuits

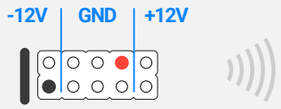
1. Set your multimeter to continuity mode (beep mode).



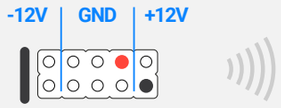
2. Place two probes on some ground pins. It should beep.



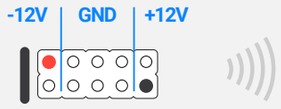
3. Keep one probe on ground and move the other one on a –12V pin. it should NOT beep.

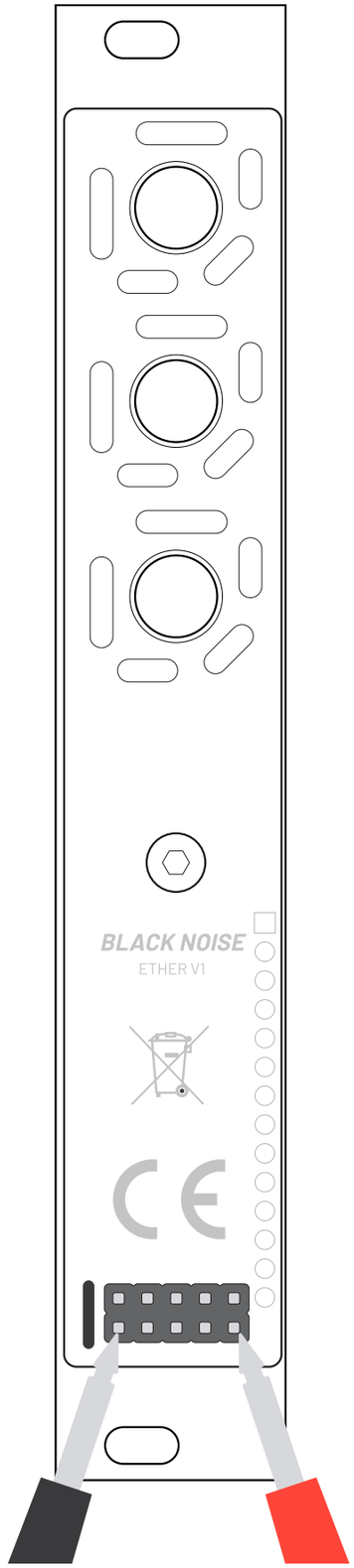


4. Keep the probe on ground and move the -12Vpin probe to a +12 V pin. it should NOT beep.



5. Keep the probe on +12 V pin and move the ground probe to a -12 V pin. it should NOT beep.





Well done!

You’ve successfully built and calibrated your Ether module!

You can now enjoy using it in your system.

For in-depth features and patch ideas, we recommend checking the full user manual available on the BLACK NOISE online ressources.

